Kwang Myung Yu

AI Engineer · Director, R&D Group, GenON

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PROFILE

Started in power plant automation and model predictive control, then developed anomaly detection, forecasting, and computer vision models for energy and manufacturing. Led RAG, agent, and Text-to-SQL services for enterprise clients, and now leads GenON's Physical AI humanoid product, GenP, from product definition to VLA training pipelines.

EXPERIENCE

GenON, R&D Group 2026 - Present
Director / AI Product Engineer
Lead the Physical AI humanoid product as Product Owner (PO), defining the product, technology tree, and commercialization roadmap, and manage client projects as Project Manager (PM). Design the integrated architecture for voice conversation, scenario execution, VLA, VLN, navigation, and motion, and build the robot control panel. Propose and run new Physical AI and agent-system projects and hire talent.

PwC Consulting, Digital & AI 2024 - 2025
Director / Senior ML Engineer
Developed generative AI services (RAG and agents), ML-based equipment diagnostics, price and demand forecasting, and control optimization. Proposed new projects, ran AI training for firm-wide and division staff, and hired talent.

MakinaRocks, AI Project Team 2023 - 2024
Part Leader / Senior ML Engineer / PO
Developed anomaly detection, control optimization, and guidance models as well as RAG-based services, and supported LLMOps and MLOps environment setup. Planned new projects, improved the traceability and reproducibility of project deliverables, researched ML technology trends, and hired talent.

Samsung Electro-Mechanics, Equipment Technology Research Institute 2021 - 2022
Principal Researcher
Built a software platform for data-driven anomaly detection and predictive maintenance and for running deep learning models. Analyzed and visualized manufacturing equipment performance and logistics data, and taught data analysis to employees.

KEPCO (Korea Electric Power Corporation), Data Science Lab 2010 - 2020
Senior Data Scientist
Researched anomaly detection algorithms, time-series forecasting, and deep learning, including sequence and generative models. Built the AI and data science roadmap, identified and planned new research projects, and planned research infrastructure and analytics environments.

POSCO E&C, Energy Business Division 2008 - 2010
Automation Design Engineer
Designed electrical and automation systems for power and energy plants, designed and built data acquisition infrastructure, developed control algorithms, and supported commissioning.

SELECTED PROJECTS

GenP Physical AI Humanoid Product 2026 - Present
GenON
Leading development of GenP, an end-to-end Physical AI humanoid product built on a common core with domain profiles for customer-specific deployments.

Docent Humanoid Robot for a Brand Experience Center Planned 2026 - 2027
GenON · Global F&B company
Delivering a humanoid docent robot that guides visitors through a brand experience center with voice conversation and autonomous navigation.

SCM Data Analysis and Work Support Service 2025
PwC Consulting · Samsung Electronics
Led an LLM service that analyzes supply chain management (SCM) data and supports planners' daily work with Text-to-SQL and agents.

Online Consultation and Product Recommendation Assistant 2024 - 2025
PwC Consulting · Samsung Electronics
Led an agent and RAG assistant that supports online customer consultation and recommends products.

SKILLS

Programming
Python for ML and services, C/C++ for control and embedded code, plus SQL and TypeScript.

AI/ML
LLM agents and RAG with LangChain/LangGraph, LiteLLM, OpenRouter, FastMCP, and Braintrust; anomaly detection, time-series forecasting, and computer vision with PyTorch and scikit-learn.

Robotics
Humanoid development on Unitree G1 and AgiBot X2 SDKs, VLA training, simulation in Isaac Lab and MuJoCo, and SLAM-based navigation.

Serving & Infrastructure
Model and LLM serving with FastAPI, vLLM, and Ollama on Docker and AWS, with MLflow, PostgreSQL, and vector databases (pgVector, Qdrant).

Domains
Energy and utilities (thermal power, renewables, ESS), petrochemical and steel processes, manufacturing, healthcare, and industrial automation (PLC, DCS).

EDUCATION

M.S. in Electrical Engineering
KAIST · 2006 - 2008

B.S., School of Electrical Engineering
Pusan National University · 1999 - 2006

AWARDS

Best People Awards
PwC Consulting · 2025

President's Award (Excellence Prize), Big Data & AI Competition
Korea East-West Power (EWP) · 2020

Merit Award (R&D)
KEPCO · 2018

Honors graduate; academic excellence scholarship
Pusan National University · 2000, 2003 - 2006

SOFTWARE

Registered 8 · Pending 2 — full list at sguys99.github.io

REPORTS

19 — full list at sguys99.github.io

Raw-Material Price Forecasting and Market Analysis Dashboard 2024
PwC Consulting · LOTTE Fine Chemical
Built a raw-material price forecasting model and market analysis dashboard to support purchasing decisions, as a PoC within the AI master plan engagement.

Green-Material Inventory Blend Optimization 2024 – 2025
PwC Consulting · LOTTE Fine Chemical
Ran a PoC that optimizes blend combinations of green-material inventory, as part of the AI master plan engagement.

OTHER PROJECTS

- 2026 Agentic RAG Search Service for National Legal Information
- 2025 Conversational Service for Power Plant Maintenance History PwC Consulting · Power generation company
- 2024 AI Master Plan and LLM Service PoC PwC Consulting · LOTTE Fine Chemical
- 2023 ML and Control Optimization for a Sinter Plant MakinaRocks · Steel maker
- 2023 Q&A Chatbot for a Manufacturing Equipment Diagnosis System MakinaRocks
- 2023 ML Models for Petrochemical Equipment Lifetime Prediction and Condition Diagnosis MakinaRocks · Petrochemical company
- 2022 Lifestyle Guide Service Application GIVita
- 2021 Visual Inspection Automation Software and ML Platform Samsung Electro-Mechanics
- 2019 Anomaly Detection from Transformer Oil DGA Data KEPCO
- 2019 Object Detection for Distribution Equipment Diagnosis KEPCO
- 2018 Power Usage Pattern Analysis of EV Charging Stations in Jeju KEPCO
- 2016 Solar Power Forecasting for Distribution Lines KEPCO
- 2015 Commissioning and Performance Evaluation System for Frequency Regulation ESS KEPCO
- 2013 Optimal Control for Power Plant Boiler Systems KEPCO

PATENTS

Registered 7 · Filed 8

- 2026 Method and system for automatically generating P&ID work reports and verifying them against the original drawings using a vision-language model Filed
- 2020 Main steam temperature control apparatus and method Registered
- 2020 Apparatus and method for diagnosing power equipment Filed
- 2019 Traffic analysis apparatus for network-based control systems and operating method thereof Registered
- 2018 Control system for a hybrid energy storage system Filed
- 2018 Power plant simulator Registered
- 2017 Superheater temperature control method Registered
- 2017 Frequency control system Filed
- 2015 Steam turbine dynamics simulator, gas turbine dynamics simulator, and turbine generator control apparatus and method using the same Filed
- 2015 Model-based predictive control apparatus and method Filed
- 2014 Operation control method for an oxy-fuel combustion boiler Registered
- 2014 Air supply apparatus and air supply method for combustion equipment Registered
- 2014 Integrated control apparatus for a supercritical Brayton cycle with a variable heat source Filed
- 2014 Interface apparatus for a process control system Registered
- 2013 Lubricant supply apparatus Filed

PAPERS

- 2026 A Novel Diagnostic Method on the Industrial Boilers Using Machine Learning: t-SNE, Auto-Encoder and LGBM Energies, vol. 19, no. 4, art. 874 · doi.org/10.3390/en19040874
- 2018 Lessons Learned from Energy Storage System Demonstrations for Primary Frequency Control KEPCO Journal on Electric Power and Energy, vol. 4, no. 2, pp. 107–114 · doi.org/10.18770/KEPCO.2018.04.02.107
- 2015 Development of a DMC Block for Use with an RCP System and its Application Journal of Institute of Control, Robotics and Systems, vol. 21, no. 9, pp. 827–835 · doi.org/10.5302/J.ICROS.2015.15.0086

BOOKS

- 2022 Learning Git & GitHub with Python Code Youngjin.com
- 2021 Git for Visual Studio Users WikiDocs

TEACHING & ADVISORY

- 2020 Visiting Researcher / Technical Advisor Department of Safety Engineering, Incheon National University
- 2026 Lectures on LLMs, agents, and Harness Engineering for undergraduate and master's researchers Engineering Education Center, Incheon National University
- 2025 AI training curriculum for consultants (design and lectures) PwC Consulting
- 2014 Hands-on course on control systems and data analysis Korea Measurement & Control Technology Education Center
- 2023 Big data analyst training program SK planet Tacademy
- 2023 Software version control with Git and GitHub (parts 1 and 2) Koscom
- 2023 Advisory Committee Member, Future Education Incheon National University
- 2019 Data science lectures for new ICT employees KEPCO Human Resources Development Institute